Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

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Abstract. This paper investigates how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, traffic flow, and waiting time. The proposed workflow integrates mobility traces, station utilization forecasts, and synthetic grid constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak grid load while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension: vehicles are no longer only energy consumers but can also feed power back, turning charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial and temporal structure of demand a central planning question. Existing siting studies, however, often rely on data or tooling that is not openly reproducible [4]. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap, and traffic is simulated microscopically in SUMO, controlled through its Python interface TraCI. From land-use points of interest (residential buildings, offices, and amenities) we derive travel demand, and we model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes with vehicle-specific access control. A custom web interface lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand and load under realistic mobility patterns; a detailed electrical model of the distribution grid (power flow, feeder and transformer constraints) as well as battery degradation and market or pricing mechanisms are beyond its scope and left to future work.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy exchange. This means that the spatial distribution of charging stations can have a significant impact on both user experience and grid load. For example, placing stations in high-traffic areas may improve accessibility but could also lead to congestion and increased waiting times. Conversely, locating stations in less busy areas might reduce queues but could limit their utilization and the potential for vehicle-to-grid (V2G) services. Understanding the spatial and temporal patterns of charging demand is therefore crucial for effective infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Given the afforementioned challenges, the problem statement of this paper can be formulated as follows: How can we determine the optimal spatial distribution of bidirectional charging stations, while regarding existing charging stations, in an urban area by integrating traffic simulation data with partially synthetic grid constraints, in order to maximize accessibility and minimize waiting times for electric vehicle users?

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. This work contains a user interface for deterministic scenario-based traffic generation along with a thorough analysis of candidate station portfolios under varying demand and grid conditions. By integrating synthetic grid constraints with mobility-based accessibility metrics, we provide a holistic framework for siting decisions that can be adapted to different urban contexts and policy objectives.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.
- A multi-criteria scoring framework for comparing candidate station portfolios under proto-realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.
- A reality-based implementation of synthetic wallbox profiles, for which no real-world data ist currently available.
- A fully working pipeline for independent scenario generation, traffic simulation, and evaluation of candidate station portfolios, which can be adapted to different urban contexts.
- A user interface for scenario-based traffic generation and evaluation of candidate station portfolios, enabling stakeholders to adjust parameters and visualize the impact of different siting strategies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 11]. Most contributions optimize either where demand is best served or how the grid copes with the additional load; bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, EV density, and adjacency to high-traffic corridors [6], which motivates deriving demand from land-use data as we do. Microscopic traffic simulators are increasingly used for this purpose: SUMO in particular provides built-in energy and charging-infrastructure models [7] and has been used to assign electric vehicles to charging stations from simulated traffic data [8]. Such approaches improve accessibility and reduce detours, but typically abstract away local constraints in the distribution grid.

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation, frequently relying on metaheuristic optimization to jointly site and size stations under electrical constraints [5]. While they improve technical feasibility, they can yield a poor user experience when travel behavior is not modeled in sufficient detail. The placement of charging stations is often motivated not only by charging demand and by economic advantages, but also by repercussions for the network [12–14].

2.3 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side, while leaving a detailed electrical model of the distribution grid to future work.

3. Methodology

Our methodology links a mesoscopic traffic simulation with a distribution-grid assessment model and a ranking module for candidate station layouts. Each scenario is evaluated over multiple demand profiles to capture variability and uncertainty.

3.1 System Overview

The workflow consists of four stages: data preprocessing, demand simulation, grid impact estimation, and multi-criteria scoring. Inputs include road-network topology, travel demand traces, candidate station coordinates, and feeder-level electrical limits. Outputs are scenario-level KPIs that quantify accessibility, utilization, and network stress.

3.2 Model Architecture

Demand prediction is represented by a spatiotemporal model that maps traffic states and contextual features to charging events. A separate grid module estimates nodal loading and transformer utilization from aggregated charging schedules. Both modules are coupled through iterative updates so that mobility and grid effects remain consistent.

The loss function is defined as:

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i(\theta)\right)^2 + \lambda\|\theta\|_2^2 \qquad (1)$$

where y_i denotes the ground-truth label and $\hat{y}_i$ the model prediction.

3.3 Training Procedure

Model parameters are trained with mini-batch optimization and early stopping based on validation loss. We perform repeated runs with different random seeds to reduce variance in performance estimates. Hyperparameters are selected using a bounded search space to balance accuracy and computational cost.

Table 1: Model configuration hyperparameters

Parameter	**Value**	**Search Space**
Learning Rate	0.001	$[10^{-4}, 10^{-2}]$
Batch Size	64	{32, 64, 128}
Hidden Layers	3	{2, 3, 4}
Dropout Rate	0.2	$[0.1, 0.5]$
Epochs	100	–

4. Experiments

We evaluate the proposed planning approach on representative urban scenarios with varying fleet sizes and charging behaviors. The experiments compare baseline siting heuristics against our integrated optimization strategy.

4.1 Experimental Setup

All experiments are executed on synchronized traffic and grid snapshots with 15-minute resolution. We run each scenario over multiple simulated days to account for temporal fluctuations in demand and network loading.

4.2 Datasets

The traffic dataset includes synthetic trip chains calibrated to observed commuting and leisure patterns. Grid data contains feeder topology, transformer ratings, and operational constraints provided at district level.

4.3 Evaluation Metrics

We report accessibility metrics (average detour time and queueing delay), utilization metrics (station occupancy and throughput), and grid metrics (peak loading and overload frequency). In addition, we compute a composite score to summarize planning trade-offs across objectives.

4.4 Results

The integrated method consistently improves accessibility while reducing critical grid stress events relative to single-domain baselines. Benefits are strongest during evening peaks, where coordinated siting shifts demand away from constrained feeders. Overall, results suggest that multi-domain optimization yields more robust infrastructure portfolios.

Figure 1: Comparison of model accuracy on the test dataset.

5. Discussion

The results highlight the importance of coupling transportation demand with local grid constraints during station planning. In particular, locations that appear optimal from a pure mobility perspective may create unacceptable stress at nearby substations.

5.1 Interpretation of Results

Our findings indicate that balanced station portfolios can improve user service levels without relying on costly grid reinforcement in every high-demand area. The strongest gains arise when moderate detour increases are traded for substantial reductions in feeder overload risk.

5.2 Limitations

This study relies on scenario assumptions for travel behavior, charging preferences, and controllability of vehicle-to-grid operation. Real-world deployment may introduce additional uncertainties, such as policy constraints and user acceptance effects, that are not fully captured in the current model.

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

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